SUNNY RISHY VARDHAN

Jersey City, NJ | +1 860 997 9603 | sunnyrishyvardhan@gmail.com | LinkedIn | GitHub

Summary

Software Engineer with an M.S. in Computer Science and strong C/C++ and Object Oriented Design skills on Linux. Solid grounding in OS concepts, data structures and real time systems with hands on embedded exposure through robotics and drone platforms.

Education

Adelphi University Aug 2024 – May 2026
Master's, Computer and Information Sciences | GPA: 3.84/4.0 *Garden City, NY*

- Inducted into Upsilon Pi Epsilon (UPE) International Honor Society for Computing and Information Disciplines
- Coursework: Advanced Operating Systems, Compiler Construction, Advanced Algorithms, Database Management Systems

National Institute of Technology, Andhra Pradesh Aug 2020 – May 2024
Bachelor's, Computer Science and Engineering | GPA: 7.4/10.0 *Andhra Pradesh, India*

- Coursework: Operating Systems, Data Structures and Algorithms, Object Oriented Programming, Computer Networks, Microprocessors and Embedded Systems

Experience

express-rate-limit (Open Source Contribution) Jun 2026 – Aug 2026
Merged Contributor | TypeScript, Node.js, Express | PR #661 *Remote*

- Implemented and shipped a `retryAfter` configuration feature across express-rate-limit's middleware and type system, landing a production PR merged after maintainer review on a package with 49.8M+ weekly npm downloads.
- Built an isolated reproduction harness with a live Dockerized Redis instance to diagnose a suspected store-recovery bug, validating real failure behavior and correcting an initial hypothesis before shipping a fix.

Adelphi University Nov 2025 – May 2026
Graduate Research Assistant in Algorithms and Systems *Garden City, NY*

- Conducted research under Dr. Sixia Chen. Designed and implemented 3 novel optimization algorithms on a 1.8M edge real world dataset improving solution quality 0.293% over the state of the art baseline with provable performance guarantees.
- Built automated benchmarking infrastructure across 100+ randomized trials using Python and shell scripting. Analyzed results to isolate bottlenecks and drive measurable improvements.

Projects

Real Time Multi Threaded Systems Engine | *C, C++17, Linux* | *GitHub*

- Built a real time multi threaded engine from scratch on Linux processing 1.4M events per second at 400ns operation latency. Applied Object Oriented Design with lock free queues, atomics and cache line aware memory layouts grounded in kernel scheduling and memory management behavior.
- Achieved 25x throughput improvement through kernel level profiling with perf, elimination of heap allocations on the critical path and microbenchmark driven validation at every step.
- Diagnosed race conditions, memory corruption and latency spikes using gdb and strace. Traced execution across user space and kernel boundaries to isolate root cause in each case.

Linux Network Service and Protocol Stack Handler | *C++17, PostgreSQL, Docker* | *GitHub*

- Designed and built a multi threaded network service on Linux using TCP and IP socket programming. Achieved under 50ms response times via async IO, connection pooling and thread safe request routing across concurrent clients.
- Debugged deadlocks and protocol level failures using gdb and strace with structured logging for full visibility into system behavior. Implemented unit and integration test coverage across all modules.
- Containerized with Docker and automated build and deployment through CI/CD pipelines for reproducible releases across environments.

High Performance Indexing and Data Structures Engine | *C++, STL* | *GitHub*

- Built an indexing and retrieval engine applying advanced data structure design. Processed 136K tokens across 1,005 documents in under 1 second at 145K tokens per second with 2.36ms average query latency.
- Reduced memory footprint 30% through container selection and allocation strategy tuning while holding sub millisecond response under concurrent load.

Skills

- **Languages:** C, C++17, Python, Bash, Java, SQL
- **Embedded and Platform:** Embedded Systems, Real Time Software, Hardware Software Integration, IoT
- **Operating Systems:** Linux, UNIX, Kernel Concepts, Process Scheduling, Memory Management, IPC, Concurrency
- **Core Fundamentals:** Object Oriented Design, Algorithms, Complexity Analysis, Compiler Construction, Software Architecture
- **Networking:** TCP, UDP, IP, Socket Programming, Computer Networks, Protocol Handling
- **Debugging and Tools:** gdb, strace, perf, Structured Logging, Root Cause Analysis, Git, Docker, CMake, CI CD Pipelines

Leadership

Women in STEM Event, Adelphi University Apr 2026
Mentor and Student Judge *Garden City, NY*

- Volunteered as mentor and student judge. Evaluated participant projects and guided aspiring students in computing and engineering.

E-Yantra Robotics Club, NIT Andhra Pradesh Aug 2023 – May 2024
Joint Secretary *Andhra Pradesh, India*

- Led 10+ hands on workshops on Robotics, IoT and Embedded Systems for 50+ students. Coordinated a robotics competition attracting 200+ participants.